Statistics

Chapter 9 — Instructor Solutions Manual

October 27, 2026 • Total Exercises: 4

PROBLEM STATEMENT

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Solution A

Step 1

Given:

$$\mu = \text{Mean} = 12.2$$
$$\sigma = \text{Standard deviation} = 2.3$$
$$n = \text{Sample size} = 12$$
$$\overline{x} = 11$$

Since the population distribution is normal, the sampling distribution of the sample mean $\overline{x}$ is also normal.

The sampling distribution of the sample mean $\overline{x}$ has mean μ and standard deviation $\dfrac{\sigma}{\sqrt{n}}$.

The z-score is the value decreased by the mean, divided by the standard deviation:

$$z = \frac{\overline{x} - \mu_{\overline{x}}}{\sigma_{\overline{x}}} = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}} = \frac{11 - 12.2}{2.3/\sqrt{12}} \approx -1.81$$

Determine the corresponding probability using the normal probability table in the appendix. $P(0 < Z < 1.81)$ is given in the row starting with 1.8 and in the column starting with .01 of the standard normal probability table in the appendix.

$$\begin{aligned} P(\overline{X} \leq 11) &= P(Z < -1.81) \\ &= P(Z > 1.81) \\ &= 1 - P(Z < 1.81) \\ &= 1 - P(Z < 0) - P(0 < Z < 1.81) \\ &= 1 - 0.5000 - 0.4649 \\ &= 0.0351 \\ &= 3.51\% \end{aligned}$$

Result

$$0.0351 = 3.51\%$$

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Solution A

Step 1

Given:

$$\begin{aligned}\overline{x} &= \text{Sample mean} = 9.3\\ s &= \text{Sample standard deviation} = 2\\ n &= \text{Sample size} = 26\\ c &= \text{Confidence level} = 90\% = 0.90\end{aligned}$$

(a) The sample mean of 9.3 years refers to the number of years that 26 residents of Conway, SC lived at their current address.

The corresponding population mean then represents the mean number of years that all residents of Conway, SC lived at their current address.

Step 2

(b) The best estimate of the population mean is the sample mean, which is 9.3 years in this case.

Step 3

(c) The standard error of the estimate is the sample standard deviation divided by the square root of the sample size.

$$SE = \frac{s}{\sqrt{n}} = \frac{2}{\sqrt{26}} \approx 0.3922$$

Thus the standard error of the estimate is 0.3922 years.

Step 4

(d) **Confidence interval**

Determine the t-value by looking in the row starting with degrees of freedom $df = n - 1 = 26 - 1 = 25$ and in the column with $c = 90\%$ in the table of the Student's T distribution:

$$t_{\alpha/2} = 1.708$$

The margin of error is then:

$$\begin{aligned} E &= t_{\alpha/2} \times \frac{s}{\sqrt{n}} \\ &= 1.708 \times \frac{2}{\sqrt{26}} \\ &\approx 0.6699 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\overline{x} - E = 9.3 - 0.6699 = 8.6301$$

$$\overline{x} + E = 9.3 + 0.6699 = 9.9699$$

We are 90% confident that the true population mean is between 8.6301 years and 9.699 years.

Result

(a) The mean number of years that all residents of Conway, SC lived at their current address.

(b) 9.3 years

(c) 0.3922 years

(d) $(8.6301, 9.9699)$

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Solution A

Step 1

Given:

$$\pi = \text{Population proportion} = 27\% = 0.27$$
$$E = \text{Margin of error} = 2\% = 0.02$$
$$c = \text{Confidence coefficient} = 98\% = 0.98$$

Formula sample size:

$$n = \frac{z^2 \pi (1-\pi)}{E^2}$$

For confidence level $1-\alpha = 0.98$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.02/2 = 0.4900$ using table the normal probability table in the appendix:

$$z = 2.33$$

Evaluate the formula for the sample size (round up to the nearest integer!):

$$n = \frac{z^2 \pi (1-\pi)}{E^2} = \frac{2.33^2 \times 0.27(1-0.27)}{0.02^2} \approx 2676$$

Thus we require a sample of size at least 2676.

Note: The solution in the back of the book appears to have rounded down to $n = 2675$ instead of rou

Result

$$n = 2676$$

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Solution A

Step 1

Given:

$$x = \text{Number of successes} = 64$$
$$n = \text{Sample size} = 100$$
$$c = \text{Confidence level} = 95\% = 0.95$$

Confidence interval

The sample proportion is the number of successes divided by the sample size:

$$\pi = \frac{x}{n} = \frac{64}{100} = 0.64$$

For confidence level $1 - \alpha = 0.95$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.05/2 = 0.4975$ using table the normal probability table in the appendix:

$$z = 1.96$$

The margin of error is then:

$$\begin{aligned} E &= z \times \sqrt{\frac{\pi(1-\pi)}{n}} \\ &= 1.96 \times \sqrt{\frac{0.64(1-0.64)}{100}} \\ &\approx 0.09408 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\pi - E = 0.64 - 0.09408 = 0.54592$$

$$\pi + E = 0.64 + 0.09408 = 0.73408$$

We are 95% confident that the true population proportion of workers that used public transportation is between 0.54592 and 0.73408.

🎯 Result

$$(0.54592, 0.73408)$$